Python Assignment
Topic-8: Control Statements (Break, Continue, Pass)

Q1. Understanding the role of break statement in Python loops.

Ans.

The break statement is used to stop a loop immediately. When Python finds a break statement inside a loop, it exits the loop even if the condition is still true. It is useful when we want to stop the loop based on a specific condition. Break helps in saving time and avoids unnecessary iterations. It is commonly used in search operations. Using break makes programs more efficient and faster.

Example:

```
for i in range(5):
    if i == 3:
        break
    print(i)
```

Q2. Understanding the role of continue statement in Python loops.

Ans.

The continue statement is used to skip the current iteration of a loop. When Python encounters continue, it skips the remaining code inside the loop for that iteration and moves to the next iteration. It is useful when we want to ignore certain values in a loop. Continue helps in controlling the flow of the loop. It makes programs cleaner and easier to manage.

Example:

```
for i in range(5):
    if i == 3:
        continue
    print(i)
```

Q3. Understanding the role of pass statement in Python.

Ans.

The pass statement is used as a placeholder in Python. It does nothing when executed. It is useful when a statement is required syntactically but no action is needed. Pass is often used in empty functions, loops, or conditions. It helps in writing code step by step without errors. It is useful during development when the logic is not yet implemented.

Example:

```
for i in range(3):
    pass
```